JavaScript

Ready to unlock web interactivity? JavaScript transforms static pages into dynamic experiences.

Created by Brendan Eich in 1995, this versatile language now powers everything from simple animations to complex web applications.

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ISTM -Web-Based Programming

What is JavaScript?

Client-Side

Runs directly in the user's browser. No server processing needed for basic interactions.

Interpreted

Code executes without compilation. Browsers read and run the code line by line.

Multi-Paradigm

Supports various programming approaches. Works with object-oriented, functional, and procedural styles.

Role of JavaScript in Web Development

Interactivity

Adds dynamic elements to web pages. Creates responsive user interfaces and experiences.

DOM Manipulation

Changes HTML and CSS on the fly. Updates content without refreshing the page.

Server Communication

Fetches data asynchronously. Powers modern single-page applications through APIs.

JavaScript vs HTML and CSS

1. **JavaScript**
   Behavior & Functionality
2. **CSS**
   Presentation & Style
3. **HTML**
   Structure & Content

These three languages work together seamlessly. HTML provides structure, CSS handles appearance, and JavaScript adds interactivity.

Together they form the cornerstone of modern web development.

Setting up JavaScript: Embedding in HTML

Inline Script

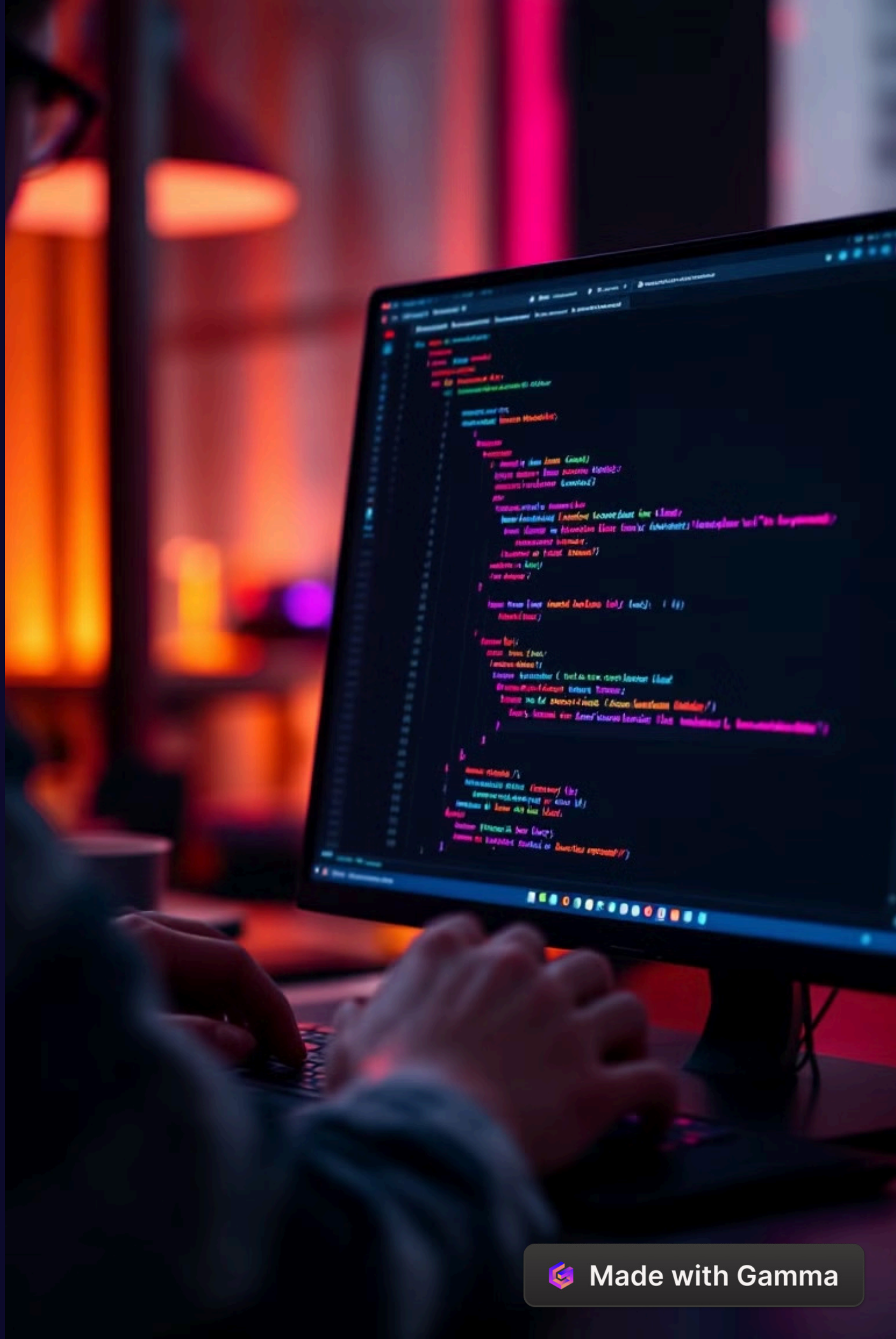
Add code directly between <script> tags in HTML. Quick for small snippets and testing.

Internal Script

Place script tags in document head or body. Keeps code separate from HTML structure.

External File

Link to separate .js files. Improves organization, caching, and reusability.



JavaScript Basics: Variables

var

Function-scoped variable declaration. Can be redeclared and updated. Legacy method but still supported.

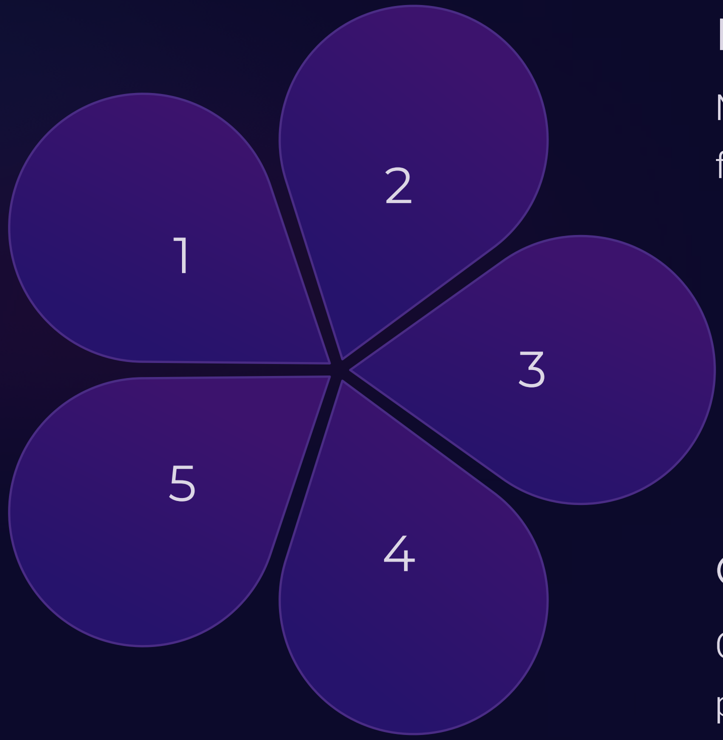
let

Block-scoped variable declaration. Can be updated but not redeclared within scope. Introduced in ES6.

const

Block-scoped constant declaration. Cannot be updated or redeclared. Used for fixed values.

JavaScript Basics: Data Types



Strings

Text values in quotes. Used for names, messages, and other textual content.

Numbers

Numeric values. Includes integers and floating-point numbers.

Booleans

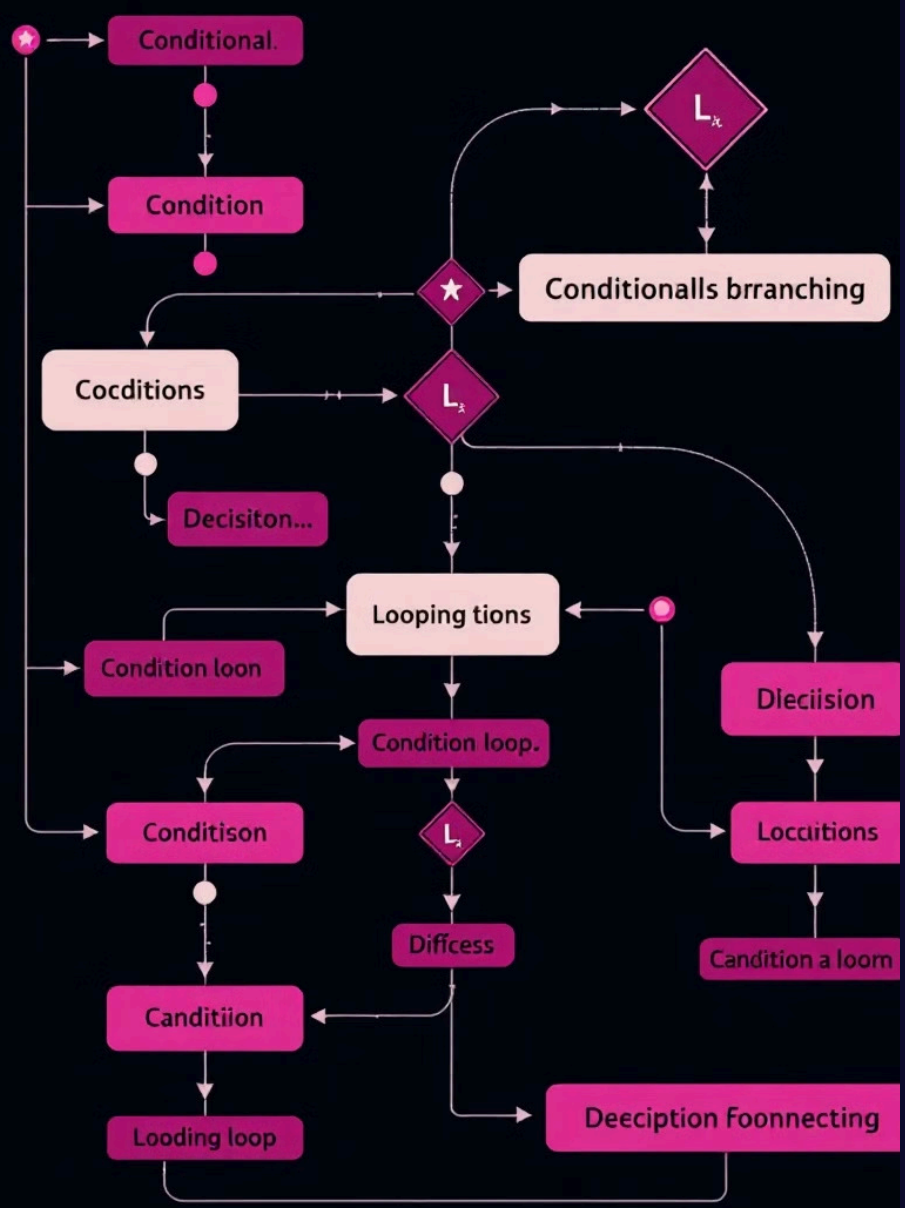
True or false values. Used for conditional logic and comparisons.

Objects

Collections of related data. Stores key-value pairs for complex structures.

Arrays

Ordered lists of values. Stores multiple items in a single variable.



Control Structures

1

Conditional Statements

Make decisions in code. Use if/else statements to execute different blocks based on conditions.

2

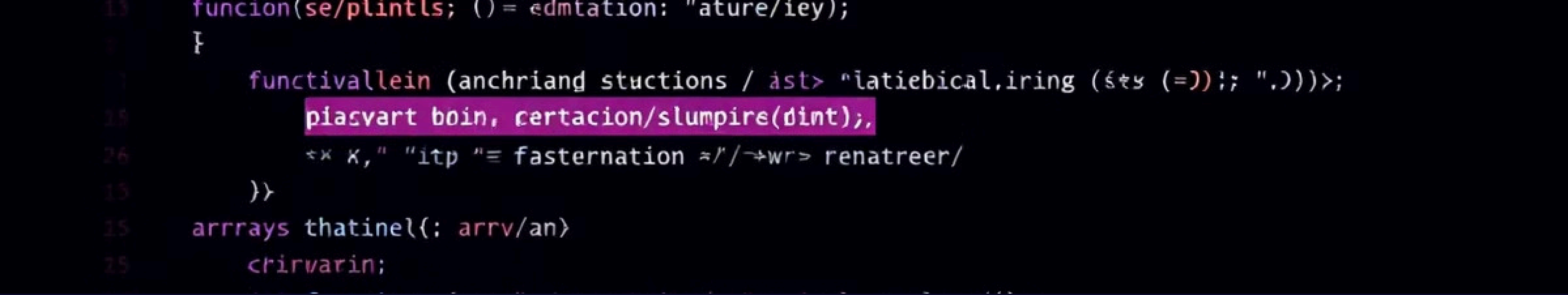
Loops

Repeat code blocks efficiently. For loops iterate a specific number of times.

3

Break & Continue

Control loop execution flow. Skip iterations or exit loops early based on conditions.



Functions and Arrays

1. **Function Declaration**
   Define reusable code blocks. Create with function keyword or arrow syntax.

2. **Parameters & Arguments**
   Pass data to functions. Input values that functions work with.

3. **Arrays**
   Store ordered collections. Access elements by index, starting at zero.

4. **Array Methods**
   Manipulate array data. Use push(), pop(), map(), and filter() for common operations.

DOM Manipulation

1

Select Elements

Target HTML components. Use getElementById() or querySelector() methods.

2

Modify Content

Change what users see. Update text and HTML with innerHTML or textContent.

3

Style Elements

Adjust appearance dynamically. Modify CSS properties through JavaScript.

4

Handle Events

Respond to user actions. Add event listeners for clicks, input, and more.